INDIVIDUAL PROJECT REPORT

LOAN EMI & ELIGIBILITY CALCULATOR

A rule-based loan screening and decision-support prototype

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EDUCATIONAL PROTOTYPE • SYNTHETIC DATA • HUMAN REVIEW REQUIRED

Executive Summary

This project develops a simple Loan EMI and Eligibility Calculator that combines a standard reducing-balance EMI calculation with a rule-based eligibility screen. The calculator accepts monthly income, existing EMI, credit score, loan amount, annual interest rate and tenure in months. It calculates the proposed EMI, total monthly EMI burden and Fixed Obligation to Income Ratio (FOIR), then applies the assigned baseline rule.

Baseline eligibility rule

An application is marked eligible only when FOIR is less than or equal to 50% AND the credit score is greater than or equal to 700. Both conditions must pass.

The supplied teaching workbook contains 500 synthetic applications and 17 fields. The project evaluates the baseline rule, checks calculations, compares 25 combinations of FOIR and credit-score thresholds, and translates the results into a short credit-policy recommendation. The analysis is descriptive: the historical 24-month default field is used to evaluate how past outcomes align with the rule; it is not treated as a prediction of future behaviour.

Problem Statement

Loan screening requires more than looking at the requested loan amount. The proposed monthly instalment must be considered alongside existing EMI obligations and monthly income, while the credit score provides a second screening signal. Without a consistent calculation method, similar applications could be assessed inconsistently and the reasons for rejection may not be transparent.

The project problem is to create a repeatable, understandable tool that calculates EMI and FOIR, applies a clearly defined eligibility rule, explains why an application passes or fails, and evaluates the rule on the provided dataset.

Project Objectives

- Calculate proposed EMI using the reducing-balance method and safely handle zero interest.
- Calculate total EMI burden and FOIR from the applicant's monthly income.
- Apply the baseline rule FOIR ≤ 50% AND credit score ≥ 700.
- Provide a clear eligibility result, condition statuses and rejection reason.
- Evaluate the rule against historical outcomes across all 500 synthetic records.
- Compare 25 threshold combinations and explain the approval-versus-risk trade-off.
- Document limitations and responsible-use safeguards for an educational prototype.

1. Dataset Description and Preparation

The analysis uses the supplied Excel workbook, primarily the Applications worksheet. The original application records are treated as source data and should remain unchanged. Calculations and rule decisions are added in separate analysis columns/sheets so the original fields can be traced back to the supplied dataset.

Field group	Fields	Purpose in analysis
Identifier	application_id	Distinguishes each application record.
Applicant profile	applicant_name, gender, age, city, city_tier, employment_type, dependents	Describes the synthetic applicant profile; not all fields are required by the baseline rule.
Income and obligations	monthly_income_inr, existing_emi_inr	Used to assess repayment burden and calculate FOIR.
Credit and loan terms	credit_score, loan_purpose, loan_amount_inr, tenure_months, interest_rate_pct, years_in_current_job	Credit score and loan terms drive eligibility calculations; purpose and employment tenure provide context.
Historical outcome	defaulted_within_24m	Used only for retrospective evaluation of the screening rule.

Dataset Summary

Item	Recorded value
Number of applications	500
Number of source fields	17
Missing values in Applications sheet	0, as recorded in the existing project checks
Historical defaults	43
Historical non-defaults	457
Data type	Synthetic educational dataset

Data Preparation Principles

- Preserve the original Applications sheet; do not overwrite source values.
- Check field names, data types, numeric values and missingness before calculation.
- Treat income, existing EMI, loan amount, tenure and interest rate as numeric inputs with valid ranges.
- Do not replace invalid or zero monthly income with an invented value; flag the calculation as invalid.
- Keep defaulted_within_24m separate from the eligibility rule so historical outcomes do not leak into the decision.

Interpretation note

The dataset is synthetic and contains only 500 records. Findings describe this teaching sample and cannot establish how a real lender's portfolio would perform.

2. Methodology and Calculation Framework

The project follows a staged workflow so that the app, spreadsheet and report use the same definitions. Each calculation is performed before the final eligibility decision, and the historical outcome is consulted only during evaluation.

Stage	Method
1. Inspect source data	Review the workbook structure, identify the Applications worksheet and confirm the fields required for the calculations.
2. Calculate proposed EMI	Use the loan principal, monthly interest rate and tenure to calculate the fixed monthly instalment.
3. Calculate total monthly EMI	Add existing EMI to the proposed EMI to estimate the applicant's total monthly EMI obligations.
4. Calculate FOIR	Divide total monthly EMI obligations by monthly income and multiply by 100.
5. Apply the baseline rule	Check FOIR and credit score separately, then combine the two conditions using AND logic.
6. Explain the result	Return eligible/not eligible and a rejection reason identifying which condition failed.
7. Evaluate historical outcomes	Compare eligibility decisions with defaulted_within_24m for the 500 records.
8. Compare alternative rules	Run all 25 combinations of five FOIR thresholds and five credit-score thresholds.
9. Verify and report	Cross-check representative calculations, test the app, reconcile counts and document limitations.

Core Formulas

Reducing-balance EMI

EMI = P × r × (1 + r)^n ÷ [(1 + r)^n − 1]

P = principal/loan amount; r = monthly interest rate as a decimal; n = tenure in months.

Monthly interest rate

r = Annual interest rate (%) ÷ 12 ÷ 100

Zero-interest case

If the interest rate is 0%, EMI = Loan Amount ÷ Tenure in months.

FOIR

FOIR (%) = [(Existing EMI + Proposed EMI) ÷ Monthly Income] × 100

The EMI formula assumes a fixed interest rate and equal monthly instalments. It does not include processing fees, insurance, taxes or other charges unless they are explicitly added to the inputs. FOIR is a screening measure, not a complete affordability assessment.

3. Eligibility Rule and Calculator Application

Baseline Rule Logic

Condition	Pass criterion	Meaning
FOIR condition	FOIR ≤ 50%	Total monthly EMI obligations are no more than half of monthly income.
Credit-score condition	Credit score ≥ 700	The score meets the minimum threshold set for this assignment.
Final decision	Both conditions pass	Eligible only if FOIR and credit score both meet the rule.

Boundary handling is important: a FOIR of exactly 50% passes the FOIR condition, and a credit score of exactly 700 passes the score condition. If one or both conditions fail, the result is not eligible and the app should identify the failed condition(s).

Application Interface

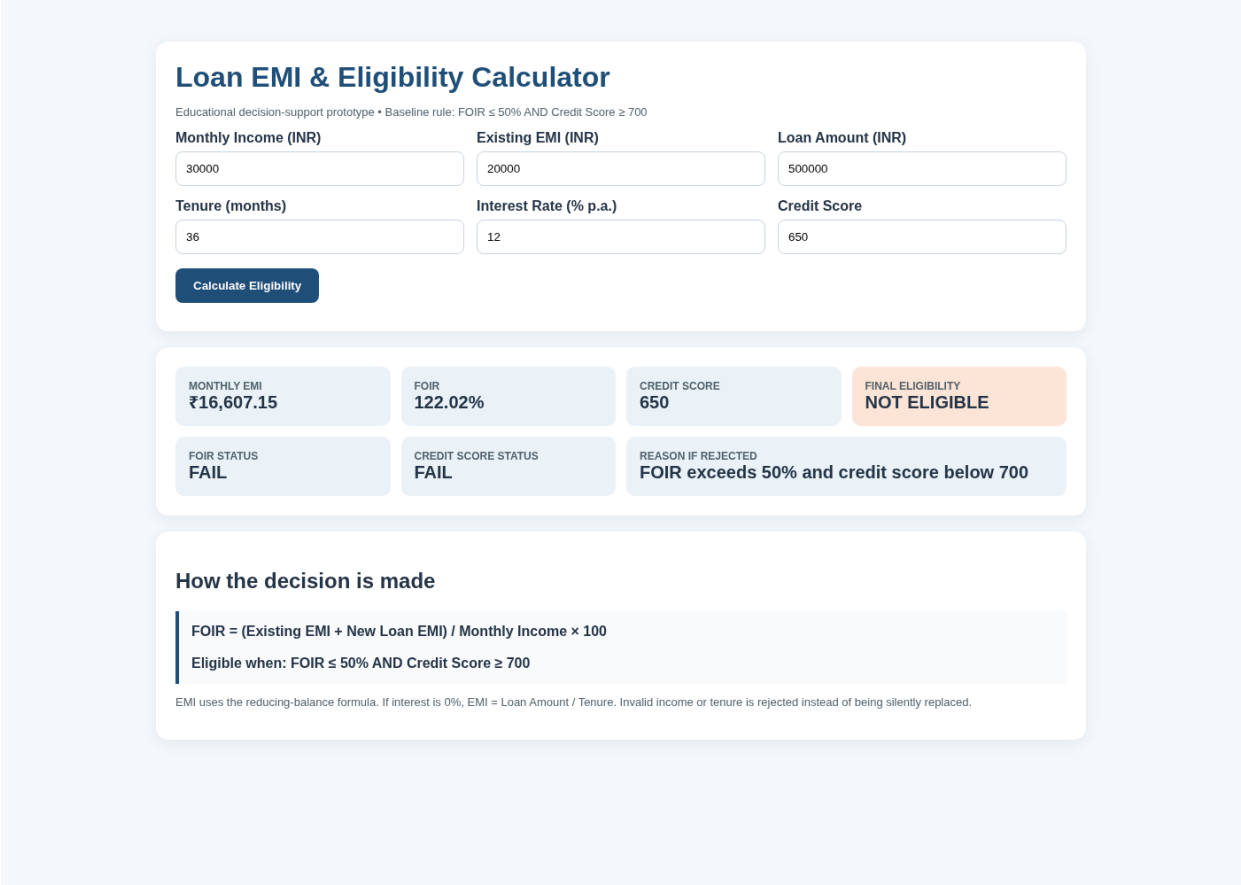


Figure 1. Calculator interface showing input fields, calculated EMI and FOIR, condition status and final decision.

The interface follows the blue-and-white visual style of the supplied application. Inputs include monthly income, existing EMI, loan amount, tenure, annual interest rate and credit score. Results include monthly EMI, FOIR, credit score, final eligibility, each condition's status and a rejection reason.

The application is provided as a self-contained HTML file that can be opened in a modern browser. It is an educational prototype and does not connect to a bank, credit bureau or live customer database.

4. Application Validation and Test Approach

Validation checks reduce the risk of producing misleading outputs from invalid inputs or incorrect formula logic. The supplied project records indicate that ten representative application records were used as test cases and marked as passing after comparison with expected values.

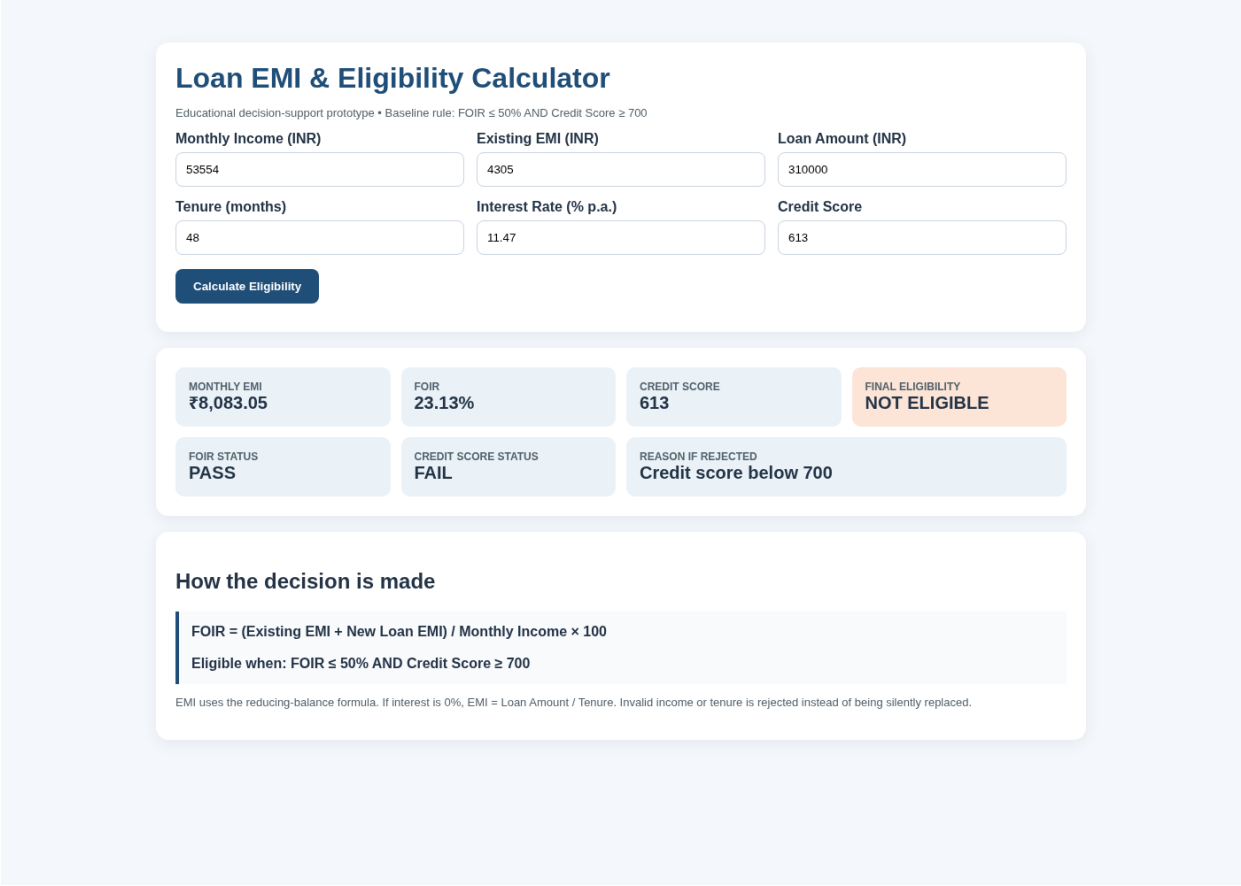


Figure 2. Calculator screenshot from a test case where FOIR passes but the credit-score condition fails.

Validation checklist

Check	Expected behaviour	Why it matters
Normal positive inputs	EMI and FOIR are calculated and displayed.	Confirms the normal calculation path.
Zero interest	EMI equals loan amount divided by tenure.	Prevents division by zero in the EMI formula.
FOIR exactly 50%	FOIR condition passes.	Confirms threshold inclusivity.
Credit score exactly 700	Score condition passes.	Confirms score threshold inclusivity.
Both conditions pass	Final decision is eligible.	Checks AND logic.
Only FOIR fails	Not eligible; FOIR reason shown.	Checks rejection explanation.
Only score fails	Not eligible; score reason shown.	Checks rejection explanation.
Both conditions fail	Not eligible; both reasons shown.	Checks combined failure logic.
Zero/invalid income	Input is rejected or marked invalid.	Avoids division by zero and misleading percentages.
Non-positive tenure or invalid values	Input validation prevents calculation.	Avoids impossible or unreliable outputs.

The existing submission reports ten application test cases as passed. For formal submission, the test evidence should remain consistent with the actual application and workbook; a report should not mark a test as passed unless it was run and verified.

5. Baseline Results and Rule Comparison

The baseline rule was applied to all 500 applications. For evaluation, the observed default flag is compared with the screening decision. A rejected applicant who later defaulted is counted as a captured default; an approved applicant who later defaulted is a default accepted by the rule.

Baseline measure	Result
Total applications	500
Eligible / approved by rule	203 (40.6%)
Not eligible / rejected by rule	297 (59.4%)
Observed defaults	43
Observed non-defaults	457
Defaults accepted (approved defaults)	9
Defaults captured by rejection	34
Good applicants rejected (non-defaults rejected)	263
Default rate among eligible	4.43%
Accuracy (supporting metric only)	45.60%

Confusion Matrix

Rule outcome	Historical default	Historical non-default
Rejected	34 (defaults captured)	263 (good applicants rejected)
Approved	9 (defaults accepted)	194 (good applicants approved)

Selected Threshold Combinations

FOIR / score	Eligible	Good rejected	Defaults accepted	Accuracy
40% / 650	216	247	6	49.40%
45% / 700	187	278	8	42.80%
50% / 700	203	263	9	45.60%
55% / 700	224	243	10	49.40%
60% / 700	244	224	11	53.00%

The 55% / 700 alternative increases approvals from 203 to 224 and reduces good applicants rejected from 263 to 243, but it also increases defaults accepted from 9 to 10. The 60% / 700 rule approves more applicants but accepts more observed defaults. These results illustrate a trade-off; they do not establish a universally optimal threshold.

6. Credit Policy: Explanation and Recommendation

Purpose of the policy

The policy translates the calculation and rule into a consistent screening process. It specifies the required inputs, the formulas used, the threshold conditions and how to explain a not-eligible result. A clear policy helps make decisions more repeatable and allows a reviewer to trace the reason for the result.

Policy rule

Primary rule for this assignment

Approve for the purposes of the calculator only when FOIR ≤ 50% AND credit score ≥ 700. Otherwise, mark the application not eligible and report the failed condition(s).

Operating steps

- Collect the six required inputs and confirm that values are valid.
- Calculate proposed EMI using the reducing-balance formula.
- Add existing EMI to proposed EMI to obtain total monthly EMI obligations.
- Calculate FOIR as total monthly EMI obligations divided by monthly income.
- Check FOIR and credit score independently against the policy thresholds.
- Display the result and a plain-language rejection reason.
- Keep the historical outcome out of the live eligibility decision; use it only for evaluation.
- For any real lending context, route the case through authorised human review and applicable lending controls.

Recommendation from this dataset

Retain the faculty-provided 50% FOIR / 700 credit-score rule as the main baseline because it directly satisfies the assigned task. The 55% / 700 option may be presented as a monitored comparison scenario when discussing higher approval volume, but it should not silently replace the baseline. In this sample, the alternative produces 21 more eligible applications and 20 fewer good applicants rejected, while accepting one additional observed default.

Policy limitations

- The results are based on synthetic data and cannot be generalised to real borrowers.
- The default flag is historical; association does not prove causation.
- The calculator does not include all affordability, legal, fraud, or regulatory checks.
- The rule is not a credit-risk model trained to predict future default.
- No threshold should be adopted for real lending without validated data, fairness review, explainability, monitoring and authorised governance.

7. Limitations, Responsible Use and Conclusion

Limitations

This project has been developed for learning and demonstration. The source dataset is synthetic, has a limited number of records, and may not represent the distribution or behaviour of real loan applicants. The chosen rule uses only FOIR and credit score. It does not include a full credit history, verified income, loan purpose-specific risk, external bureau variables, affordability stress tests, fraud checks or regulatory assessments.

The historical default indicator is used only to evaluate how the rule classifies the provided records. It should not be used as an input to the calculator's eligibility decision because doing so would use the outcome being evaluated to make the decision. The observed performance metrics are descriptive and should not be interpreted as guaranteed future performance.

Responsible AI and data handling

- Use only the supplied synthetic teaching dataset; do not enter real customer personal or financial data.
- Review calculations and generated code independently instead of assuming AI-assisted output is correct.
- Keep the original dataset unchanged and maintain a traceable record of formulas and assumptions.
- Explain the reason for each not-eligible result in plain language.
- Require human oversight, fairness evaluation, privacy safeguards and regulatory review before any real-world use.

Conclusion

The project delivers a rule-based Loan EMI and Eligibility Calculator, a documented calculation method, application screenshots, a dataset review, a baseline evaluation and a comparison of alternative thresholds. The baseline rule marked 203 of 500 applications as eligible and accepted 9 of the 43 historical defaults. It also rejected 263 applicants who did not default, demonstrating why accuracy alone is insufficient and why both risk capture and good-applicant rejection should be reviewed.

The main learning is that a calculator is not complete merely because it returns a number. Reliable use also requires input validation, transparent rules, test cases, reconciliation with the source data and careful interpretation of results. The final recommendation is to preserve the assigned 50% / 700 rule as the baseline for this educational task and clearly label any alternative as a comparison, not as a production lending policy.